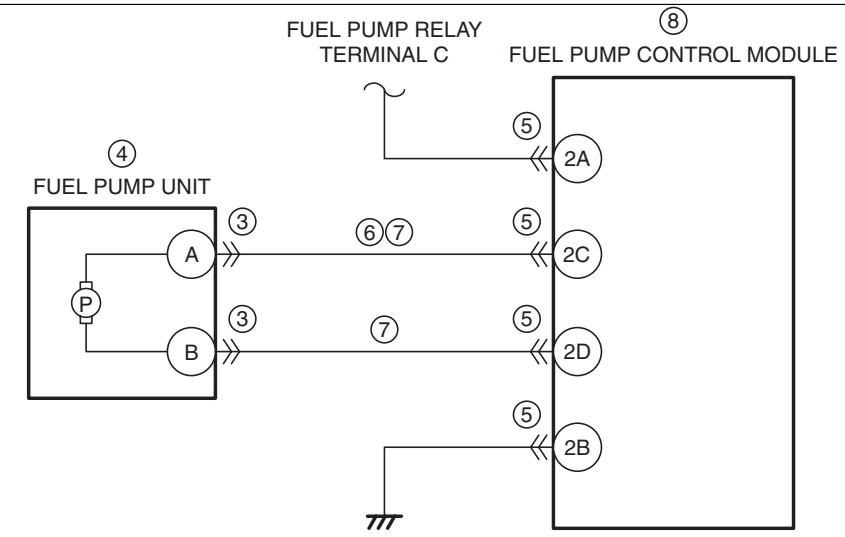
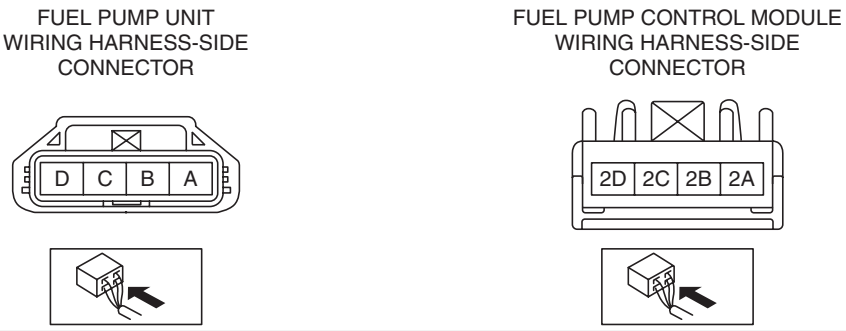
DTC P0628:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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DTC P0628:00	Fuel pump control module circuit low input
DETECTION CONDITION	• When the PCM outputs a duty signal to the fuel pump control module, the difference in voltage between fuel pump unit terminals A and B is **less than 1.5 V** for a continuous **5 s**. **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light does not illuminate. • FREEZE FRAME DATA is not available. • Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• Fuel pump unit connector or terminals malfunction • Fuel pump unit malfunction • Fuel pump control module connector or terminals malfunction • Short to ground in wiring harness between fuel pump unit terminal A and fuel pump control module terminal 2C • Open circuit in wiring harness between the following terminals: — Fuel pump unit terminal A—Fuel pump control module terminal 2C — Fuel pump unit terminal B—Fuel pump control module terminal 2D • Fuel pump control module malfunction • PCM malfunction





Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT FUEL PUMP UNIT CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the fuel pump unit connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
4	**INSPECT FUEL PUMP UNIT** • Inspect the fuel pump unit. (See FUEL PUMP UNIT INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Replace the fuel pump unit, then go to Step 9. (See FUEL PUMP UNIT REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
5	**INSPECT FUEL PUMP CONTROL MODULE CONNECTOR CONDITION** • Disconnect the fuel pump control module connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
6	**INSPECT FUEL PUMP UNIT CONTROL CIRCUIT FOR SHORT TO GROUND** • Verify that the fuel pump unit and fuel pump control module connectors are disconnected. • Inspect for continuity between fuel pump unit terminal A (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between fuel pump unit terminal A and fuel pump control module terminal 2C. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 9.
		No	Go to the next step.
7	**INSPECT FUEL PUMP UNIT CIRCUIT FOR OPEN CIRCUIT** • Verify that the fuel pump unit and fuel pump control module connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — Fuel pump unit terminal A—Fuel pump control module terminal 2C — Fuel pump unit terminal B—Fuel pump control module terminal 2D • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Fuel pump unit terminal A—Fuel pump control module terminal 2C • Fuel pump unit terminal B—Fuel pump control module terminal 2D **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.

STEP	INSPECTION		ACTION
8	**INSPECT FUEL PUMP CONTROL MODULE** • Inspect the fuel pump control module. (See FUEL PUMP CONTROL MODULE INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Replace the fuel pump control module, then go to the next step. (See FUEL PUMP CONTROL MODULE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
9	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Perform the KOEO or KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
10	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.